

MacFree — iOS Builder

Build and ship your Unity iOS game to Apple TestFlight from a **Windows or Linux PC**. No Mac. No Xcode on your machine. No command-line crypto.

This guide assumes you have **never** used an Apple Developer account, GitHub, or Unity Cloud before. Every step is spelled out. Read Part 1 once; after that, shipping a build is three clicks.

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1. What MacFree does & how it works

Apple normally requires a Mac to build and upload an iOS app. MacFree removes that requirement. It runs entirely inside the Unity Editor on Windows or Linux and hands the Mac-only parts (compiling the Xcode project, code-signing, and uploading) to a cloud machine that Apple and the build service already provide.

The end result: your game appears in **TestFlight** — Apple's app for distributing test builds — where you and your testers can install it on real iPhones and iPads.

1.1 The idea in one picture

1. **CONNECT** — you paste a few keys once so MacFree can talk to Apple, GitHub, and Unity on your behalf.
2. **SET UP EVERYTHING** — one click. MacFree creates your signing certificate, provisioning profile, and the cloud build configuration for you. You never touch `openssl` or a Mac.
3. **BUILD & SHIP** — one click. A cloud machine builds your game, signs it, and uploads it to TestFlight. About 15–25 minutes later, it's ready to test.

Good news

You only do Part 3 (accounts & keys) **once**. After that, every new build is just CONNECT is already filled → BUILD & SHIP.

1.2 Which engine should I pick?

MacFree can drive two different cloud build services. You choose one in the **Engine** dropdown. Both reach the same TestFlight result.

Engine	Best for	What it needs
GitHub Actions (recommended)	Fully automated, no website dashboards to click through. Free monthly build minutes for public repos and a generous free tier otherwise.	A free GitHub account + one access token, and your Unity login.
Unity Build Automation	People already using Unity Cloud / Unity DevOps.	A Unity Cloud project with Build Automation enabled, plus <i>one</i> manual step: connecting your repository in the Unity dashboard.

This guide's default

We walk through the **GitHub Actions** engine because it needs zero dashboard clicking. Where the Build Automation engine differs, we call it out. If you're new, use GitHub Actions.

2. What you need before you start

Gather these once. Section 3 shows exactly how to get each one.

- A paid **Apple Developer Program** account (US\$99/year) — required by Apple to sign and upload any app.
- A Windows or Linux PC with **Unity 6** installed, including the **iOS Build Support** module.
- A free **GitHub** account (for the GitHub Actions engine).
- Your **Unity account** email, password, and serial (Personal is fine — it's free).
- About **30–40 minutes** for the one-time setup.

Add the iOS Build Support module

MacFree needs Unity's iOS module to build. In **Unity Hub** → **Installs** → the gear next to your Unity version → **Add Modules** → tick **iOS Build Support** → Install. Skipping this causes build errors later.

3. One-time setup: your accounts & keys

Do these in order. Keep a notepad open — you'll collect a handful of values (a downloaded file and some IDs) that you paste into MacFree in Step 1.

3.1 Apple Developer Program membership

Apple requires a paid membership to distribute any app, even a test build.

1. Go to developer.apple.com/programs/enroll.
2. Sign in with your Apple ID (or create one). Turn on two-factor authentication if asked.
3. Enroll as an **Individual** (simplest) or **Organization**, pay the US\$99 annual fee, and wait for the confirmation email. Approval is usually minutes to a day.

3.2 App Store Connect API key (.p8, Key ID, Issuer ID)

This key lets MacFree talk to Apple without your password. You'll get **three** things: a downloaded `.p8` file, a **Key ID**, and an **Issuer ID**.

1. Go to [App Store Connect → Users and Access → Integrations → App Store Connect API](#).
2. Click the blue + (Generate API Key). Name it `MacFree`.
3. For **Access**, choose **App Manager** (this role is required — a lower role can't upload builds).
4. Click **Generate**. Now:
 - **Download the `.p8` file** and keep it somewhere safe. *Apple lets you download it only once.*
 - Copy the **Key ID** (a 10-character code next to the key).
 - Copy the **Issuer ID** (a long code shown at the top of the same page — it's the same for all your keys).

Save the .p8 now

If you lose it, you can't re-download it — you'd have to revoke it and make a new one. Store it with your other MacFree values.

3.3 Your Apple Team ID

Your Team ID is a 10-character code identifying your developer account. (*Needed for the GitHub Actions engine.*)

1. Go to developer.apple.com/account.
2. Scroll to **Membership details**. Copy the value labelled **Team ID** (looks like `226HK7FQMK`).

3.4 Create your app record (and pick a Bundle ID)

A **Bundle ID** is your app's unique name in reverse-website form, e.g. `com.yourstudio.yourgame`. It must match in three places: your Unity project, your Apple app record, and MacFree (which reads it from Unity automatically).

1. **Set it in Unity first:** `Edit → Project Settings → Player → Other Settings → Identification → Bundle Identifier`. Use your own unique value like `com.yourstudio.yourgame`. *Do not leave Unity's default* (`com.DefaultCompany....`) — Apple rejects it.
2. **Create the matching app on Apple:** go to [App Store Connect → Apps](#) → the + → **New App**. Choose iOS, give it a name, pick your Bundle ID from the list (if it's not listed, register it once at [Certificates, Identifiers & Profiles → Identifiers](#)), and create.

Why this matters

MacFree uploads a build *to a specific app record*. If the app record doesn't exist yet, the upload has nowhere to land.

3.5 A GitHub account & access token

GitHub runs the free cloud machine that builds your game. You need an account and one **token** (a long password-like string that lets MacFree push your project and start builds).

1. Create a free account at github.com/signup if you don't have one.
2. Inside MacFree (Step 1), click the link "**Create a classic token with 'repo' + 'workflow' scopes**". It opens GitHub's token page with the correct boxes **already ticked**.
3. On that page: set an **Expiration** (e.g. 90 days or No expiration), leave the ticked `repo` and `workflow` boxes as they are, scroll down, and click **Generate token**.
4. Copy the token (starts with `ghp_`) and paste it into MacFree's **GitHub token** box. *GitHub shows it only once.*

Both boxes matter

The `workflow` scope is what lets MacFree install the build recipe. If you ever make the token by hand, make sure **both** `repo` and `workflow` are ticked, or the setup will fail to push.

3.6 Your Unity login details (GitHub Actions engine)

The cloud machine needs to activate Unity to build your game, so it needs your Unity **serial**, **email**, and **password**. Unity **Personal** (free) works.

- **Serial:** in MacFree's **Unity serial** field, click **Detect** — it reads the serial this editor is already activated with. (Personal serials start with `F` and are fully supported.)
- **Email & password:** the same ones you use to sign in to the Unity Hub.

Kept private

MacFree stores these as **encrypted GitHub secrets** on your own repository. They are only used by the build machine to activate Unity.

3.7 Unity Cloud & Build Automation (Build Automation engine only)

Skip this section if you're using the GitHub Actions engine.

1. Enable Build Automation (the free tier is enough) at unity.com/products/unity-devops.
2. Link your project to Unity Cloud in the editor: **Edit → Project Settings → Services** → create or select a cloud project. This fills in your **Org ID** and **Project ID** automatically.
3. Get your **Build Automation API key** from the Unity Cloud dashboard → Build Automation → Settings.
4. **The one manual step:** in the Unity Cloud dashboard → Build Automation → **Config** → **Source Control**, connect your GitHub repository (Unity has no way to automate this). MacFree does everything else.

4. Installing MacFree

1. Open your Unity project (Unity 6, with the iOS Build Support module — see Part 2).
2. Import the MacFree package: `Assets → Import Package → Custom Package...` → choose `MacFree-iOS-Builder.unitypackage` → **Import**.
3. Open the tool: `Tools → MacFree iOS Builder`. A window appears with three sections: **CONNECT**, **SET UP**, and **SHIP**.

5. Step 1 — CONNECT (fill in your keys)

This is where you paste everything you gathered in Part 3. Do it once; MacFree remembers it.

1. **Engine** — choose **GitHub Actions (no dashboard)** (recommended) or **Unity Build Automation**.
2. **.p8 key file** — click **Pick...** and select the `.p8` file you downloaded in 3.2.
3. **Key ID** and **Issuer ID** — paste the two values from 3.2. (The link "Where do I get these?" opens the right Apple page.)
4. **GitHub token** — paste the token from 3.5. Use the "Create a classic token..." link if you don't have one yet.

If you chose the GitHub Actions engine, also fill:

- **Apple Team ID** — from 3.3.
- **Unity serial** — click **Detect**.
- **Unity email** and **Unity password** — your Unity Hub login.

If you chose the Build Automation engine, instead fill:

- **Unity Org ID** and **Unity Project ID** — fill automatically once the project is linked to Unity Cloud (3.7). If empty, click **Open Services settings**.
- **Build Automation API key** — from 3.7.

You're ready when...

the CONNECT header shows a check mark (✓). That means every required field is filled.

6. Step 2 — SET UP EVERYTHING

Click the big **SET UP EVERYTHING** button. In one pass MacFree:

- registers your Bundle ID with Apple,
- creates your iOS **Distribution certificate** and **provisioning profile** (the digital IDs Apple requires to sign an app),
- stores your keys as **encrypted secrets** on your build service,
- and (GitHub Actions) creates a private repository, pushes your project, and installs the build recipe.

A short progress line shows what it's doing. When it finishes, the SET UP header turns to a check mark. This takes roughly 1–2 minutes.

"Reset setup state" — leave it alone

The small *Reset setup state (advanced)* link clears all progress and, on the next setup, creates a **new** Apple certificate. Apple limits how many you can have, so only use it if support tells you to.

7. Step 3 — BUILD & SHIP TO TESTFLIGHT

Click **BUILD & SHIP TO TESTFLIGHT**. A cloud machine now:

1. builds your Unity game for iOS,
2. compiles and code-signs the Xcode project,
3. uploads the finished app to TestFlight.

This typically takes **15–25 minutes**. You can close the window; the build continues in the cloud. When it's done, the build appears in [App Store Connect](#) → [your app](#) → [TestFlight](#) with the status **Complete**.

First upload of a new app

Apple may show "*Missing Compliance*" next to the build. Click **Manage** and answer the encryption question (most games select "No" / uses only standard encryption) to make it testable. MacFree can set this automatically for future builds.

8. The Preflight checklist explained

The bottom of the MacFree window shows a checklist that catches problems *before* you spend build time. Green checks are good; a red mark is something to fix.

Item	What it means
Bundle ID set	Your app has a real Bundle ID (not Unity's default). Fix in Player Settings (3.4).
App icon 1024 opaque	Apple requires a fully opaque 1024×1024 icon. If red, click the Fix icon button MacFree offers.
Git repository detected	Your project is in a Git repo (MacFree creates one for you on first setup).
Hook / files committed to git	The generated build files are committed. If red, MacFree commits them during setup.
UserSettings ignored by git	Your private keys folder is excluded from Git so it's never uploaded. MacFree adds this rule automatically.

9. Inviting testers on TestFlight

Once a build shows **Complete** in TestFlight:

1. In App Store Connect → your app → **TestFlight**, create a testing group and add tester emails, *or* add yourself under **Internal Testing**.
2. Testers install the free **TestFlight** app from the App Store, open the invite email, and tap **Install**.

10. Troubleshooting

MacFree translates most errors into a plain-English hint right in the window. Common ones:

Message you see	What to do
"...without workflow scope" / push refused	Your GitHub token is missing the workflow permission. Open your token on GitHub, tick workflow , click Update token (the value stays the same), then run SET UP again. (3.5)
"no space left on device" during the cloud build	Handled automatically — MacFree's build recipe frees runner disk space before building. If you see it, make sure you're on the latest MacFree and re-run.

Bundle ID "is not available" / rejected	You're still using Unity's default Bundle ID, or someone else took it. Set your own unique one in Player Settings and run setup again. (3.4)
"must be built with the iOS 26 SDK or later" (Apple)	Apple raised its minimum Xcode/SDK. MacFree's recipe automatically selects the newest Xcode on the build machine — update MacFree and re-ship.
Icon rejected (ITMS-91111)	Use the Preflight Fix icon button — Apple requires a fully opaque 1024×1024 icon.
Build finished but nothing on TestFlight	Open the build log from the error panel; the upload section names the exact reason (usually a missing app record or icon).
Your organization "hasn't enabled Build Automation"	(Build Automation engine) Turn it on in Unity Cloud → DevOps → Build Automation (free tier exists), then re-ship.
Duplicate BouncyCastle.Crypto.dll	Your project already includes BouncyCastle from another asset. Delete whichever copy you're not using and re-import.

11. Frequently asked questions

Do I really never need a Mac? Correct. Every Mac-only step runs on a cloud machine.

Is Unity Personal (free) enough? Yes, for the GitHub Actions engine. You still need the paid Apple Developer Program — that's Apple's rule, not MacFree's.

How much does the cloud build cost? GitHub Actions includes free minutes each month; typical projects fit comfortably. Heavy use may incur GitHub's standard per-minute charges.

Can I change my game and re-ship? Yes — just click BUILD & SHIP again. MacFree automatically increases the build number so Apple accepts it.

Where do I put the app on the App Store (not just TestFlight)? TestFlight is for testing. When you're ready, submit for review from App Store Connect → your app → Distribution.

12. Where your credentials are stored (security)

- Your keys and passwords are saved in your project's `UserSettings/MacFree/` folder, which MacFree keeps **out of Git** so they're never uploaded with your code.
- On the build service, secrets are stored **encrypted** (GitHub encrypted secrets / Build Automation environment variables) and are only used by the build machine.
- Treat your `.p8` file, GitHub token, and Unity password like passwords. If any leaks, revoke it at its source (Apple, GitHub, Unity) and run setup again.

MacFree — iOS Builder · Ship Unity iOS to TestFlight from Windows or Linux. No Mac, no openssl. · This document is included with the asset for offline reference.